

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Tutorial Sheet: Unit 2 – Sessions 1, 2 & 3 (Sections A & B)

■ Section A – Very Short Answers

1. Four main functional units: Input Unit, Output Unit, Memory Unit, and CPU (Arithmetic Logic Unit + Control Unit).
2. Control Unit: coordinates and controls all other units — it fetches and decodes instructions and issues timing/control signals that execute them.
3. Bus: a shared communication pathway (set of lines) that transfers data, addresses and control signals between CPU, memory and I/O devices.
4. ALU: the Arithmetic Logic Unit — the part of the CPU that performs arithmetic (add, subtract) and logic (AND, OR, compare) operations.
5. Address bus is used to transfer memory addresses.

■ Section B – Short Answers

1. Data Bus carries actual data between units (bidirectional); Address Bus carries the location/address of the data (unidirectional, CPU → memory); Control Bus carries command and timing signals such as Read, Write and Interrupt.
2. Program Counter: holds the address of the next instruction; during fetch its value goes to the MAR, then it auto-increments (or is set by a branch) so instructions execute in sequence.
3. Steps for Load R2, LOC: (i) PC → MAR, instruction fetched to IR, PC incremented; (ii) instruction decoded; (iii) address field LOC → MAR, memory read; (iv) data → MDR; (v) MDR → R2.
4. Limitations of single bus: only one transfer at a time (serialized), bus contention slows the system, performance limited by bus speed, a bus fault can halt the whole system.
5. Advantages: simple/low cost, easy to connect devices. Disadvantages: slower due to sharing, not suitable for high-performance systems with many transfers.